



United International University

B.Sc. in Computer Science & Engineering (CSE)

CSE 4889: Machine Learning

Midterm Exam : Fall 2025 Time: 1 Hour 30 Minutes Marks: 30

Any evidence of plagiarism or copying will be punishable according to the proctorial rules of UIU
Answer all of the following questions sequentially.

1. The customer data points are given in Table 1. Illustrate two iterations of the process [6]
to cluster the data points using modified k-means clustering with cluster $k=2$.

Table: 1

Customer ID	Annual spending(tk)	No. of purchase	Avg. Rating
301	500	12	4.5
302	2000	25	4.8
303	300	10	3.5
304	2400	30	5.0

2. Analyze ensemble learnings such as Bagging, Boosting, and Stacking, on the [6]
following dataset using base models and distinguishing models.

Animals	Country	Weight	Length
Cat	UK	3750	181
Cat	UK	3800	186
Kangaroo	USA	3250	187
Kangaroo	USA	3675	198
Horse	KSA	6000	220
Horse	KSA	4750	215

3. A research team is working on soft clustering of the sales data in Table 2. Calculate [8]
 $U^{(1)}_{22}$ where cluster number, fuzziness coefficient, and termination criterion are $C=2$,
 $m=2$, and $\epsilon=0.31$, respectively.

Table 2

Sales Data	Sales (x)	Profits (y)
P1	4	3
P2	6	7
P3	3	5
P4	4	5

4. You are working with a research team on a semi-supervised learning project. The [3+5+2] dataset includes both labeled and unlabeled samples as follows:

Type	Input (p)	Output (y)
Labeled	$p_1=[2,3]$	$y_1=0.75$
Labeled	$p_2=[3,2]$	$y_2=0.32$
Unlabeled	$p_3=[2,0]$	-----
Unlabeled	$p_4=[-3,1]$	-----
Unlabeled	$p_5=[4,2]$	-----
Unlabeled	$p_6=[-2,-1]$	-----

The initial model parameters are:

- Weight vector: $\mathbf{w} = [0.35, -0.15]$
 - Bias: $b = 0.45$
 - Learning rate: $\eta = 0.05$
 - Confidence threshold: $\tau = 0.64$
- (i) Predict which unlabeled samples qualify for labeled data based on τ .
- (ii) Using both labeled and pseudo-labeled samples, compute the updated **weights** ($\mathbf{w}_{\text{new}}$) and **bias** (b_{new}) after one iteration of learning.
- (iii) Interpret the result and discuss how the confidence threshold influences learning.